Appendix C: Full model output

Model selection

Loading the data

```
# working directory
setwd("PATH/")

# load data
data <- read.csv('3 - SPR main/analysis/readingtimes_88pp_residuals_manual-outliers-REVIEWS.c
sv', header = TRUE)
stiminfo <- read.csv('5 - metadata/frequency_length_position.csv', header = TRUE)

# set factors
data$agreement <- factor(data$agreement, levels=c('correct','incorrect'))
data$surprisal <- factor(data$surprisal, levels=c('low','high'))
data$correct_number <- factor(data$correct_number, levels=c("singular", "plural"))
data$set_number <- factor(data$set_number)

fmla <- 'logRTresidual ~ agreement * surprisal_value + agreement * correct_number +
                              (1 + agreement * surprisal_value + agreement * correct_number |
UserId) +
                              (1 + agreement * surprisal_value | set_number )'

fixed <-  'logRTresidual ~ agreement * surprisal_value + agreement * correct_number'

fixed_effects <- list('logRTresidual ~ agreement * surprisal_value + agreement * correct_numb
er', # 1
                      'logRTresidual ~ surprisal_value + agreement * correct_number',
# 2
                      'logRTresidual ~ agreement * correct_number',                         #
3
                      'logRTresidual ~ agreement * surprisal_value + correct_number',
# 4
                      'logRTresidual ~ agreement * surprisal_value',
# 5
                      'logRTresidual ~ surprisal_value + agreement + correct_number',
# 6
                      'logRTresidual ~ agreement + correct_number',                         #
7
                      'logRTresidual ~ surprisal_value + correct_number',
# 8
                      'logRTresidual ~ surprisal_value + agreement',
# 9
                      'logRTresidual ~ surprisal_value',
# 10
                      'logRTresidual ~ agreement',                                          #
11
                      'logRTresidual ~ correct_number',                                     #
12
                      'logRTresidual ~ ')
# 13

participant_effects <- c('(1 + agreement * surprisal_value + agreement * correct_number | Use
rId)',
                         '(1 + agreement * surprisal_value + correct_number | UserId)',
                         '(1 + agreement * correct_number + surprisal_value | UserId)',
                         '(1 + agreement + correct_number + surprisal_value | UserId)',
```

```
                         '(1 + agreement * correct_number | UserId)',
                         '(1 + agreement + correct_number | UserId)',
                         '(1 + agreement * surprisal_value | UserId)',
                         '(1 + agreement + surprisal_value | UserId)',
                         '(1 + surprisal_value + correct_number | UserId)',
                         '(1 + agreement | UserId)',
                         '(1 + surprisal_value | UserId)',
                         '(1 + correct_number | UserId)',
                         '(1 | UserId)'
                         )

item_effects <- c('(1 + agreement * surprisal_value | set_number )',
                  '(1 + agreement + surprisal_value | set_number )',
                  '(1 + surprisal_value | set_number )',
                  '(1 + agreement | set_number )',
                  '(1 | set_number)'
                  )

# the random effects formulas
tmp_fmlas <- list()
for (randi in 1:length(item_effects)) {
  for (randp in 1:length(participant_effects)) {

    tmp_fmla <- sprintf('%s + %s + %s', fixed,  item_effects[[randi]], participant_effects[[r
andp]])
    tmp_fmlas <- c(tmp_fmlas, tmp_fmla)

  }
}
```

Helper functions

```
n_permutations <- 1000
seed <- 42
set.seed(seed)

# formula
permute_lmer <- function(model_orig,
                         data,
                         n_permutations,
                         group_var = "UserId",
                         response_var = "logRTresidual",
                         REML=FALSE,
                         verbose=TRUE){

  formula <- formula(model_orig)
  coef_names <- names(fixef(model_orig))
  obs_coefs <- fixef(model_orig)

  # storage matrix (rows = perms, cols = fixed effects)
  perm_matrix <- matrix(NA, nrow = n_permutations, ncol = length(coef_names))
  colnames(perm_matrix) <- coef_names

  for (i in 1:n_permutations) {
    # shuffle labels
    data_perm <- data %>%
      group_by(.data[[group_var]]) %>%
        mutate(
          !!response_var := sample(.data[[response_var]])
        ) %>%
        ungroup()
    # Fit model to permuted data
    tryCatch({
      perm_model <- lmer(formula, data = data_perm, control=lmerControl(optimizer="nloptwra
p", optCtrl=list(maxfun=1e6)), REML = REML)
      perm_matrix[i, ] <- fixef(perm_model)
    }, error = function(e) {
      # Skip failed fits
    })

    if (verbose && i %% 50 == 0) {
      message(sprintf("Completed %d / %d permutations", i, n_permutations))
    }

  }

  perm_matrix <- perm_matrix[complete.cases(perm_matrix), , drop = FALSE]

  p_values <- sapply(seq_along(obs_coefs), function(j) {
    mean(abs(perm_matrix[, j]) >= abs(obs_coefs[j]))
  })
  names(p_values) <- coef_names

  return(list(
    observed_coefs = obs_coefs,
```

```
    permuted_coefs = perm_matrix,
    p_values = p_values
  ))

}

plot_permutation_densities <- function(result_list) {

  permuted_df <- as.data.frame(result_list$permuted_coefs)
  observed <- result_list$observed_coefs
  pvals <- result_list$p_values

  coef_names <- setdiff(names(permuted_df), "(Intercept)")

  plots <- lapply(coef_names, function(coef_name) {
    ggplot(permuted_df, aes(x = .data[[coef_name]])) +
      geom_density(fill = "skyblue", alpha = 0.5) +
      geom_vline(xintercept = observed[coef_name], color = "red", linewidth = 1) +
      labs(
        title = sprintf("%s\n(p = %.3f)", coef_name, pvals[coef_name]),
        x = "Coefficient value",
        y = "Density"
      ) +
      theme_minimal(base_size=10) +
      theme(plot.title = element_text(size = 10, hjust = 0.5))  # Center and shrink subplot t
itles
  })

    # Combine all plots into one figure
  combined_plot <- wrap_plots(plots, ncol = length(plots)) +
    plot_annotation(title = "Permutation Test Distributions for Fixed Effects",
      theme = theme(plot.title = element_text(hjust = 0.5, size = 14))
    )

  return(combined_plot)
}
```

```
count_random <- function(mdl) {

  vc_list <- VarCorr(mdl)

  # Count variance components
  sum(sapply(vc_list, function(v) {
    n_std  <- length(attr(v, "stddev"))
    n_corr <- if (!is.null(attr(v, "correlation"))) {
                length(attr(v, "correlation"))
              } else 0L
    n_std + n_corr
  }))
}
```

```
fit_random <- function(tmp_fmla, data) {
  model <- lmer(tmp_fmla, data, control=lmerControl(optimizer="nloptwrap", optCtrl=list(maxfu
n=2e5)), REML=FALSE)

}

fit_fixed <- function(tmp_fixed, random, data) {
  tmp_fmla <- eval(parse(text=paste(tmp_fixed, " + ", random)))
  model <- lmer(tmp_fmla, data, control=lmerControl(optimizer="nloptwrap", optCtrl=list(maxfu
n=1e6)), REML=FALSE)

}
```

Random effects - estimation of all models

Word index 5 (target)

```
models5 <- lapply(tmp_fmlas, fit_random, subset(data, word_index==5))
saveRDS(models5, file='PATH/3 - SPR main/analysis/models5.Rdata')
```

Word index 6 (spill-over 1)

```
models6 <- lapply(tmp_fmlas, fit_random, subset(data, word_index==6))
saveRDS(models6, file='PATH/3 - SPR main/analysis/models6.Rdata')
```

Word index 6 (spill-over 2)

```
models7 <- lapply(tmp_fmlas, fit_random, subset(data, word_index==7))
saveRDS(models7, file='PATH/3 - SPR main/analysis/models7.Rdata')
```

Word index 6 (spill-over 3)

```
models8 <- lapply(tmp_fmlas, fit_random, subset(data, word_index==8))
saveRDS(models8, file='PATH/3 - SPR main/analysis/models8.Rdata')
```

Evaluating the random models, fixed effects

estimation & model selection

Word index 5 (target)

```
models5 <- readRDS('PATH/3 - SPR main/analysis/models5.Rdata')

good_models5 <- list()
for (mdli in 1:length(models5)) {
  mdl <- models5[[mdli]]
  if ((isSingular(mdl) == FALSE)&(mdl@optinfo$conv$opt == 0L)&(is.null(mdl@optinfo$conv$lme4
$code))) {
    good_models5 <- append(good_models5, mdl)
  }
}

# now we sort them for number of variables
# and their AIC -- we choose the model with the *highest number of variables* and *lowest AIC
* combination
rand_counts <- vapply(good_models5, count_random, integer(1))
AIC_vals <- sapply(good_models5, AIC)
nvar = order(-rand_counts)
aics = order(AIC_vals, decreasing=FALSE)

# now we take the sum of the position of each model
var_ix = match(1:length(good_models5), nvar)
aic_ix = match(1:length(good_models5), aics)

score <- var_ix + aic_ix

sorted_models5 <- good_models5[order(score)]

# we pick the largest one
start_model5 <- sorted_models5[[1]]
print(formula(start_model5))
```

```
## logRTresidual ~ agreement * surprisal_value + agreement * correct_number +
##     (1 + agreement | set_number) + (1 + agreement | UserId)
## <environment: 0x000002731ebb7eb0>
```

```r
# now we reduce the fixed effects - STEP keeps reducing random effects too
# so we go back to manual selection
fixed5 <- lapply(fixed_effects, fit_fixed, "(1 + agreement | set_number) + (1 + agreement | U
serId)", subset(data, word_index==5))

# save the fixed models
saveRDS(fixed5, file='PATH/3 - SPR main/analysis/fixed5.Rdata')

final_models5 <- list()
for (mdli in 1:length(fixed5)) {
  mdl <- fixed5[[mdli]]
  if ((isSingular(mdl) == FALSE)&(mdl@optinfo$conv$opt == 0L)&(is.null(mdl@optinfo$conv$lme4
$code))) {
    final_models5 <- append(final_models5, mdl)
  }
}

fixed_aics = sapply(final_models5, AIC)
best_model5 = final_models5[order(fixed_aics, decreasing=FALSE)][[1]]
print(formula(best_model5))
```

```
## logRTresidual ~ surprisal_value + agreement * correct_number +
##     (1 + agreement | set_number) + (1 + agreement | UserId)
## <environment: 0x0000027323ca7958>
```

```r
summary(best_model5)
```

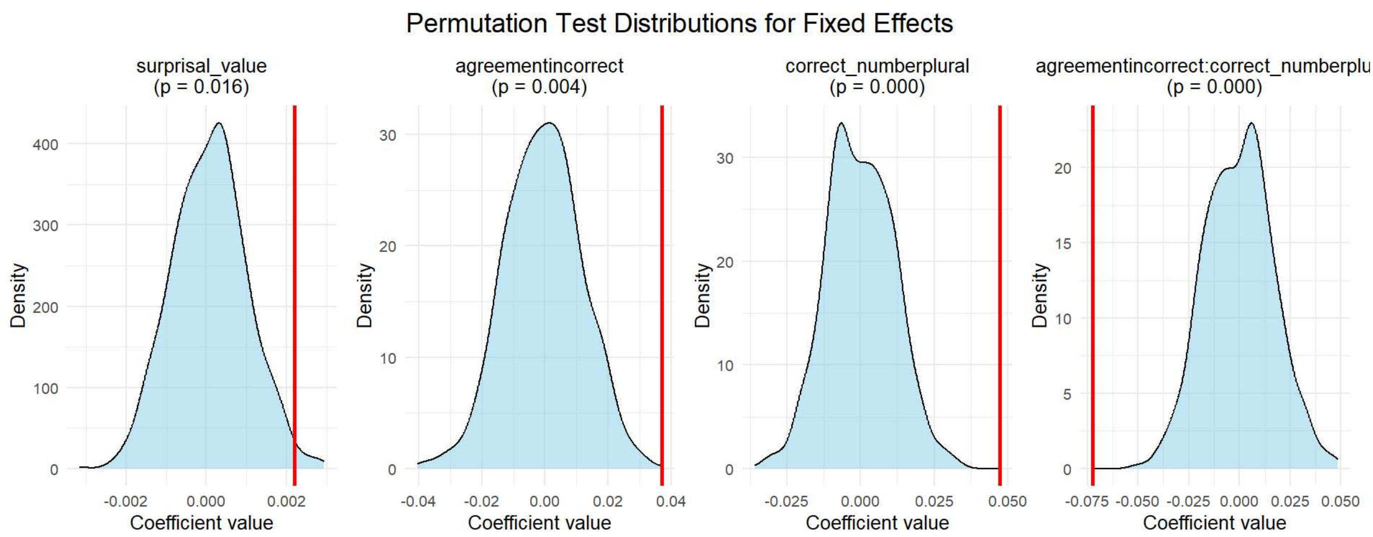
```
## Linear mixed model fit by maximum likelihood . t-tests use Satterthwaite's
##   method [lmerModLmerTest]
## Formula: tmp_fmla
##    Data: data
## Control: lmerControl(optimizer = "nloptwrap", optCtrl = list(maxfun = 1e+06))
## 
##      AIC      BIC   logLik deviance df.resid 
##    -37.3     36.5     30.7    -61.3     3453 
## 
## Scaled residuals: 
##     Min      1Q  Median      3Q     Max 
## -4.6726 -0.5927 -0.0891  0.4775  5.1373 
## 
## Random effects:
##  Groups     Name               Variance  Std.Dev. Corr 
##  UserId     (Intercept)        0.0037090 0.06090       
##             agreementincorrect 0.0021392 0.04625  -0.60
##  set_number (Intercept)        0.0019801 0.04450       
##             agreementincorrect 0.0009439 0.03072  0.23 
##  Residual                      0.0544022 0.23324       
## Number of obs: 3465, groups:  UserId, 88; set_number, 40
## 
## Fixed effects:
##                                        Estimate Std. Error         df
## (Intercept)                          -1.114e-01  1.970e-02  1.766e+02
## surprisal_value                       2.193e-03  9.992e-04  3.124e+03
## agreementincorrect                    3.716e-02  1.453e-02  4.739e+01
## correct_numberplural                  4.732e-02  1.802e-02  3.990e+01
## agreementincorrect:correct_numberplural -7.236e-02  1.863e-02  3.936e+01
##                                      t value Pr(>|t|)    
## (Intercept)                           -5.652 6.27e-08 ***
## surprisal_value                        2.195 0.028255 *  
## agreementincorrect                     2.558 0.013782 *  
## correct_numberplural                   2.626 0.012191 *  
## agreementincorrect:correct_numberplural  -3.883 0.000384 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
## 
## Correlation of Fixed Effects:
##             (Intr) srprs_ agrmnt crrct_
## surprisl_vl -0.672                     
## agrmntncrrc -0.149 -0.139              
## crrct_nmbrp -0.502  0.033  0.187       
## agrmntncr:_  0.144 -0.011 -0.673 -0.283
```

```
saveRDS(best_model5, file='PATH/3 - SPR main/analysis/best_model5.Rdata')
```

Permutation test

```
best_model5 <- readRDS('PATH/3 - SPR main/analysis/best_model5.Rdata')
perm5 <- permute_lmer(best_model5,
                      subset(data, word_index==5),
                      n_permutations,
                      verbose=TRUE)
saveRDS(perm5, file='PATH/3 - SPR main/analysis/perm55.Rdata')
```

```
plots5 <- plot_permutation_densities(perm5)
print(plots5)
```




```
print(perm5$p_values)
```

```
##                               (Intercept)                           surprisal_value
##                                     0.000                                     0.016
##                        agreementincorrect                      correct_numberplural
##                                     0.004                                     0.000
## agreementincorrect:correct_numberplural
##                                     0.000
```

Simple effects

```
# agreement by number
mdl5_agr_by_num <- emmeans(best_model5, specs='agreement', by='correct_number')

# number by agreement
mdl5_num_by_agr <- emmeans(best_model5, specs='correct_number', by='agreement')

# get results without adjustment
mdl5_agr_by_num_raw <- summary(pairs(mdl5_agr_by_num), adjust='none')
mdl5_num_by_agr_raw <- summary(pairs(mdl5_num_by_agr), adjust='none')

# extract pvalues
all_pvals <- c(mdl5_agr_by_num_raw$p.value, mdl5_num_by_agr_raw$p.value)
adj_pvals <- p.adjust(all_pvals, method='bonferroni')

# extract pvalues
mdl5_res_pair <- rbind(mdl5_agr_by_num_raw, mdl5_num_by_agr_raw)
mdl5_res_pair$adj.p.value <- adj_pvals
summary(mdl5_res_pair)
```

```
##  correct_number agreement contrast            estimate     SE   df t.ratio
##  singular       .         correct - incorrect  -0.0372 0.0145 47.4  -2.558
##  plural         .         correct - incorrect   0.0352 0.0139 47.7   2.530
##  .              correct   singular - plural    -0.0473 0.0180 39.9  -2.626
##  .              incorrect singular - plural     0.0250 0.0220 39.1   1.141
##  p.value adj.p.value
##   0.0138      0.0551
##   0.0148      0.0591
##   0.0122      0.0488
##   0.2610      1.0000
##
## Degrees-of-freedom method: satterthwaite
```

Word index 6 (spill-over 1)

```
models6 <- readRDS('PATH/3 - SPR main/analysis/models6.Rdata')

good_models6 <- list()
for (mdli in 1:length(models6)) {
  mdl <- models6[[mdli]]
  if ((isSingular(mdl) == FALSE)&(mdl@optinfo$conv$opt == 0L)&(is.null(mdl@optinfo$conv$lme4
$code))) {
    good_models6 <- append(good_models6, mdl)
  }
}

# now we sort them
rand_counts <- vapply(good_models6, count_random, integer(1))
AIC_vals <- sapply(good_models6, AIC)
nvar = order(-rand_counts)
aics = order(AIC_vals, decreasing=FALSE)

# now we take the sum of the position of each model
var_ix = match(1:length(good_models6), nvar)
aic_ix = match(1:length(good_models6), aics)
score <- var_ix + aic_ix

sorted_models6 <- good_models6[order(score)]

# we pick the best/largest one
start_model6 <- sorted_models6[[1]]
print(formula(start_model6))
```

```
## logRTresidual ~ agreement * surprisal_value + agreement * correct_number +
##     (1 | set_number) + (1 + agreement * correct_number | UserId)
## <environment: 0x000002731e2bd008>
```

```r
# now we reduce the fixed effects - STEP keeps reducing random effects too
# so we go back to manual selection
fixed6 <- lapply(fixed_effects, fit_fixed, "(1 | set_number) + (1 + agreement * correct_numbe
r | UserId)", subset(data, word_index==6))

# save the fixed models
saveRDS(fixed6, file='PATH/3 - SPR main/analysis/fixed6.Rdata')

# now we check whether they converged
final_models6 <- list()
for (mdli in 1:length(fixed6)) {

  mdl <- fixed6[[mdli]]

  if ((isSingular(mdl) == FALSE)&(mdl@optinfo$conv$opt == 0L)&(is.null(mdl@optinfo$conv$lme4
$code))) {
    final_models6 <- append(final_models6, mdl)
  }
}

# they did, so we pick the model with the highest AIC
fixed_aics = sapply(final_models6, AIC)
best_model6 = final_models6[order(fixed_aics, decreasing=FALSE)][[1]]
print(formula(best_model6))
```

```
## logRTresidual ~ agreement * surprisal_value + agreement * correct_number +
##     (1 | set_number) + (1 + agreement * correct_number | UserId)
## <environment: 0x00000273234b0ca8>
```

```r
summary(best_model6)
```

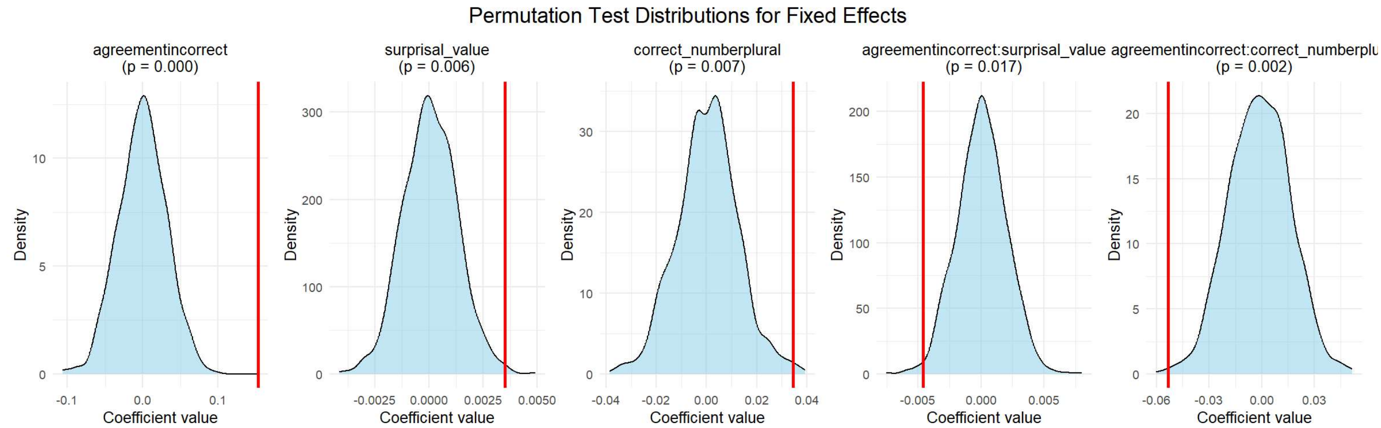
```
## Linear mixed model fit by maximum likelihood . t-tests use Satterthwaite's
##   method [lmerModLmerTest]
## Formula: tmp_fmla
##    Data: data
## Control: lmerControl(optimizer = "nloptwrap", optCtrl = list(maxfun = 1e+06))
##
##      AIC      BIC   logLik deviance df.resid
##    218.0    328.6    -91.0    182.0     3430
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -3.1829 -0.6034 -0.0662  0.5069  5.1077
##
## Random effects:
##  Groups     Name                                 Variance  Std.Dev. Corr
##  UserId     (Intercept)                          0.0005287 0.02299
##             agreementincorrect                   0.0080555 0.08975   0.65
##             correct_numberplural                 0.0004108 0.02027   0.94
##             agreementincorrect:correct_numberplural 0.0048195 0.06942  -0.35
##  set_number (Intercept)                          0.0028916 0.05377
##  Residual                                        0.0577811 0.24038
##
##
##
##   0.61
##  -0.66 -0.57
##
##
## Number of obs: 3448, groups:  UserId, 88; set_number, 40
##
## Fixed effects:
##                                           Estimate Std. Error         df
## (Intercept)                             -8.936e-02  2.286e-02  2.784e+02
## agreementincorrect                       1.533e-01  3.243e-02  1.387e+03
## surprisal_value                          3.496e-03  1.293e-03  3.201e+03
## correct_numberplural                     3.454e-02  2.068e-02  5.620e+01
## agreementincorrect:surprisal_value      -4.622e-03  1.970e-03  3.273e+03
## agreementincorrect:correct_numberplural -5.327e-02  1.804e-02  1.204e+02
##                                         t value Pr(>|t|)
## (Intercept)                              -3.910 0.000116 ***
## agreementincorrect                        4.727 2.51e-06 ***
## surprisal_value                           2.703 0.006906 **
## correct_numberplural                      1.670 0.100440
## agreementincorrect:surprisal_value       -2.346 0.019040 *
## agreementincorrect:correct_numberplural  -2.954 0.003778 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##                 (Intr) agrmnt srprs_ crrct_ agrmntncrrct:s_
## agrmntncrrc     -0.434
## surprisl_vl     -0.751  0.479
## crrct_nmbrp     -0.487  0.141  0.038
## agrmntncrrct:s_  0.453 -0.880 -0.603 -0.023
## agrmntncrrct:c_  0.183 -0.364 -0.040 -0.382  0.049
```

```
saveRDS(best_model6, file='PATH/3 - SPR main/analysis/best_model6.Rdata')
```

Permutation test

```
# load the best model
best_model6 <- readRDS('PATH/3 - SPR main/analysis/best_model6.Rdata')
perm6 <- permute_lmer(best_model6,
                      subset(data, word_index==6),
                      n_permutations,
                      verbose=TRUE)
saveRDS(perm6, file='PATH/3 - SPR main/analysis/perm6.Rdata')
```

```
plots6 <- plot_permutation_densities(perm6)
print(plots6)
```




```
print(perm6$p_values)
```

```
##                          (Intercept)                   agreementincorrect
##                                0.000                                0.000
##                      surprisal_value                 correct_numberplural
##                                0.006                                0.007
##   agreementincorrect:surprisal_value agreementincorrect:correct_numberplural
##                                0.017                                0.002
```

Simple effects

```r
# agreement by number
mdl6_agr_by_num <- emmeans(best_model6, specs='agreement', by='correct_number')

# number by agreement
mdl6_num_by_agr <- emmeans(best_model6, specs='correct_number', by='agreement')

# surprisal by agreement
mdl6_sur_by_agr <- emtrends(best_model6, var='surprisal_value', specs='agreement')

# get results without adjustment
mdl6_agr_by_num_raw <- summary(pairs(mdl6_agr_by_num), adjust='none')
mdl6_num_by_agr_raw <- summary(pairs(mdl6_num_by_agr), adjust='none')
mdl6_sur_by_agr_raw <- summary(mdl6_sur_by_agr, infer = c(FALSE,TRUE))

# extract pvalues
all_pvals <- c(mdl6_agr_by_num_raw$p.value, mdl6_num_by_agr_raw$p.value, mdl6_sur_by_agr_raw
$p.value) #,  mdl6_agr_by_sur_raw$p.value
adj_pvals <- p.adjust(all_pvals, method='bonferroni')

# extract pvalues
mdl6_res_pair <- rbind(mdl6_agr_by_num_raw, mdl6_num_by_agr_raw, mdl6_sur_by_agr_raw) #mdl6_a
gr_by_sur_raw
mdl6_res_pair$adj.p.value <- adj_pvals
summary(mdl6_res_pair)
```

```
##  correct_number agreement contrast             estimate      SE     df t.ratio
##  singular       .         correct - incorrect  -0.0885 0.01543  106.1  -5.737
##  plural         .         correct - incorrect  -0.0353 0.01365   94.4  -2.584
##  .              correct   singular - plural    -0.0345 0.02068   56.2  -1.670
##  .              incorrect singular - plural     0.0187 0.02164   61.0   0.866
##  .              correct   .                     nonEst 0.00129 3200.9   2.703
##  .              incorrect .                     nonEst 0.00157 3220.4  -0.715
##  p.value surprisal_value.trend adj.p.value
##   <.0001                    NA    6.00e-07
##   0.0113                    NA    6.78e-02
##   0.1004                    NA    6.03e-01
##   0.3901                    NA    1.00e+00
##   0.0069               0.00350    4.14e-02
##   0.4748              -0.00113    1.00e+00
##
## Degrees-of-freedom method: satterthwaite
## The following messages apply only to some rows:
## * Results are averaged over the levels of: correct_number
```

Word index 7 (spill-over 2)

```
models7 <- readRDS('PATH/3 - SPR main/analysis/models7.Rdata')

good_models7 <- list()
for (mdli in 1:length(models7)) {
  mdl <- models7[[mdli]]
  if ((isSingular(mdl) == FALSE)&(mdl@optinfo$conv$opt == 0L)&(is.null(mdl@optinfo$conv$lme4
$code))) {
    good_models7 <- append(good_models7, mdl)
  }

}

# now we sort them
rand_counts <- vapply(good_models7, count_random, integer(1))
AIC_vals <- sapply(good_models7, AIC)
nvar = order(-rand_counts)
aics = order(AIC_vals, decreasing=FALSE)

# now we take the sum of the position of each model
var_ix = match(1:length(good_models7), nvar)
aic_ix = match(1:length(good_models7), aics)
score <- var_ix + aic_ix

sorted_models7 <- good_models7[order(score)]

# we pick the best/largest one
start_model7 <- sorted_models7[[1]]
print(formula(start_model7))
```

```
## logRTresidual ~ agreement * surprisal_value + agreement * correct_number +
##     (1 + agreement | set_number) + (1 + correct_number | UserId)
## <environment: 0x000002736d429db8>
```

```
# now we reduce the fixed effects - STEP keeps reducing random effects too
# so we go back to manual selection
fixed7 <- lapply(fixed_effects, fit_fixed, "(1 + agreement | set_number) + (1 + correct_numbe
r | UserId)", subset(data, word_index==7))

# save the fixed models
saveRDS(fixed7, file='PATH/3 - SPR main/analysis/fixed7.Rdata')

# now we check whether they converged
final_models7 <- list()
for (mdli in 1:length(fixed7)) {
  mdl <- fixed7[[mdli]]
  if ((isSingular(mdl) == FALSE)&(mdl@optinfo$conv$opt == 0L)&(is.null(mdl@optinfo$conv$lme4
$code))) {
    final_models7 <- append(final_models7, mdl)
  }
}

# there was only one model left (code not necessary)
fixed_aics = sapply(final_models7, AIC)
best_model7 = final_models7[order(fixed_aics, decreasing=FALSE)][[1]]
print(formula(best_model7))
```

```
## logRTresidual ~ surprisal_value + agreement + (1 + agreement |
##     set_number) + (1 + correct_number | UserId)
## <environment: 0x000002733afb4490>
```

```
summary(best_model7)
```

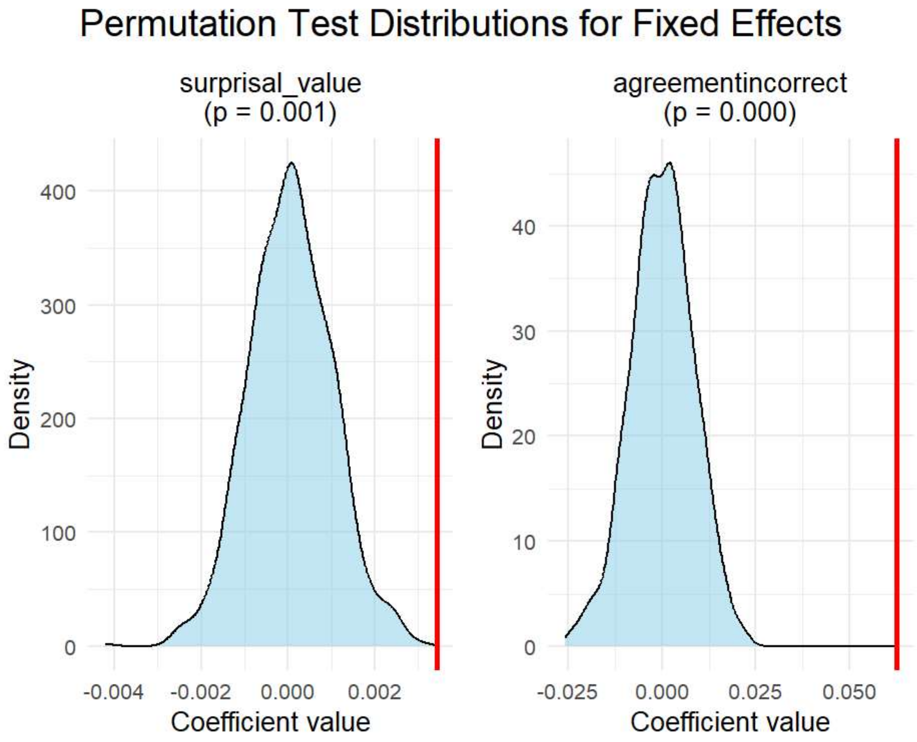
```
## Linear mixed model fit by maximum likelihood . t-tests use Satterthwaite's
##   method [lmerModLmerTest]
## Formula: tmp_fmla
##    Data: data
## Control: lmerControl(optimizer = "nloptwrap", optCtrl = list(maxfun = 1e+06))
##
##      AIC      BIC   logLik deviance df.resid
##   -123.4    -62.0     71.7   -143.4     3448
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -3.3834 -0.6049 -0.0668  0.5134  4.2551
##
## Random effects:
##  Groups     Name                 Variance  Std.Dev. Corr
##  UserId     (Intercept)          0.0025297 0.05030
##             correct_numberplural 0.0001353 0.01163  -0.52
##  set_number (Intercept)          0.0021657 0.04654
##             agreementincorrect   0.0020553 0.04534  -0.23
##  Residual                        0.0534699 0.23124
## Number of obs: 3458, groups:  UserId, 88; set_number, 40
##
## Fixed effects:
##                      Estimate Std. Error         df t value Pr(>|t|)
## (Intercept)        -6.373e-02  1.655e-02  3.216e+02  -3.851 0.000142 ***
## surprisal_value     3.428e-03  9.879e-04  3.068e+03   3.470 0.000528 ***
## agreementincorrect  6.261e-02  1.085e-02  4.260e+01   5.770 8.12e-07 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr) srprs_
## surprisl_vl -0.773
## agrmntncrrc -0.089 -0.194
```

```
saveRDS(best_model7, file='PATH/3 - SPR main/analysis/best_model7.Rdata')
```

Permutation test

```
best_model7 <- readRDS('PATH/3 - SPR main/analysis/best_model7.Rdata')
perm7 <- permute_lmer(best_model7,
                      subset(data, word_index==7),
                      n_permutations,
                      verbose=TRUE)
saveRDS(perm7, file='PATH/3 - SPR main/analysis/perm7.Rdata')
```

```
plots7 <- plot_permutation_densities(perm7)
print(plots7)
```




```
print(perm7$p_values)
```

```
##        (Intercept)    surprisal_value agreementincorrect
##              0.001              0.001              0.000
```

Word index 8 (spill-over 3)

```
models8 <- readRDS('PATH/3 - SPR main/analysis/models8.Rdata')

good_models8 <- list()
for (mdli in 1:length(models8)) {
  mdl <- models8[[mdli]]
  if ((isSingular(mdl) == FALSE)&(mdl@optinfo$conv$opt == 0L)&(is.null(mdl@optinfo$conv$lme4
$code))) {
    good_models8 <- append(good_models8, mdl)
  }
}

# now we sort them
rand_counts <- vapply(good_models8, count_random, integer(1))
AIC_vals <- sapply(good_models8, AIC)
nvar = order(-rand_counts)
aics = order(AIC_vals, decreasing=FALSE)

# now we take the sum of the position of each model
var_ix = match(1:length(good_models8), nvar)
aic_ix = match(1:length(good_models8), aics)
score <- var_ix + aic_ix

sorted_models8 <- good_models8[order(score)]

# we pick the best/largest one
start_model8 <- sorted_models8[[1]]
print(formula(start_model8))
```

```
## logRTresidual ~ agreement * surprisal_value + agreement * correct_number +
##     (1 + agreement | set_number) + (1 + agreement | UserId)
## <environment: 0x0000027365f1d190>
```

```r
# now we reduce the fixed effects - STEP keeps reducing random effects too
# so we go back to manual selection
fixed8 <- lapply(fixed_effects, fit_fixed, "  (1 + agreement | set_number) + (1 + agreement |
UserId)", subset(data, word_index==8))

# save the fixed models
saveRDS(fixed8, file='PATH/3 - SPR main/analysis/fixed8.Rdata')

# now we check whether they converged
final_models8 <- list()
for (mdli in 1:length(fixed8)) {

  mdl <- fixed8[[mdli]]

  if ((isSingular(mdl) == FALSE)&(mdl@optinfo$conv$opt == 0L)&(is.null(mdl@optinfo$conv$lme4
$code))) {
    final_models8 <- append(final_models8, mdl)
  }
}

# they did, so we pick the model with the highest AIC
fixed_aics = sapply(final_models8, AIC)
best_model8 = final_models8[order(fixed_aics, decreasing=FALSE)][[1]]
print(formula(best_model8))
```

```
## logRTresidual ~ surprisal_value + agreement + (1 + agreement |
##     set_number) + (1 + agreement | UserId)
## <environment: 0x0000027329f12518>
```

```r
summary(best_model8)
```

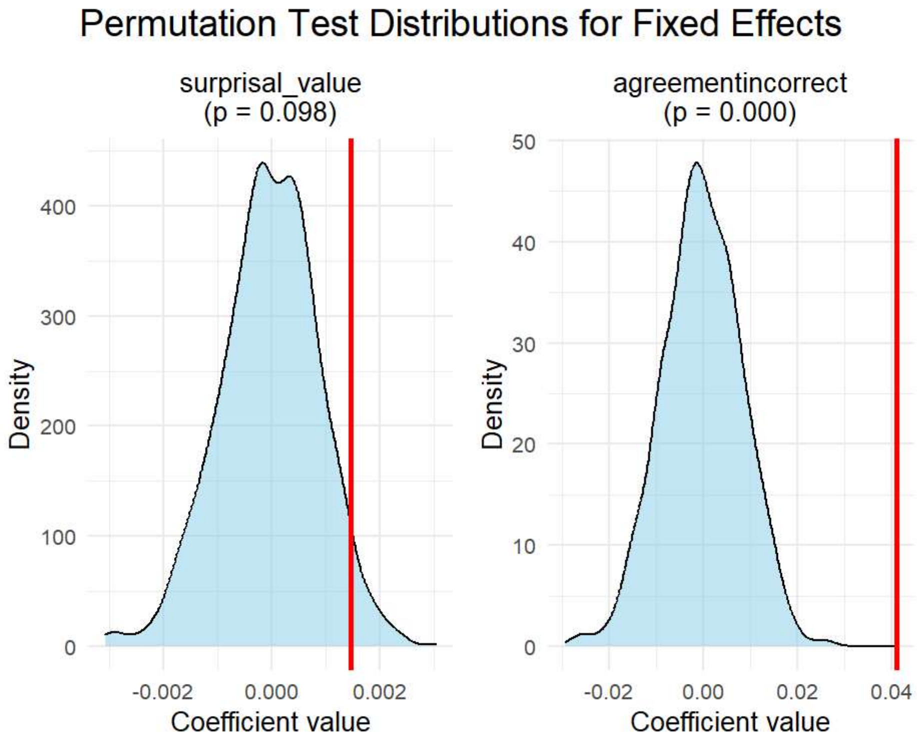
```
## Linear mixed model fit by maximum likelihood . t-tests use Satterthwaite's
##   method [lmerModLmerTest]
## Formula: tmp_fmla
##    Data: data
## Control: lmerControl(optimizer = "nloptwrap", optCtrl = list(maxfun = 1e+06))
##
##      AIC      BIC   logLik deviance df.resid
##   -344.4   -282.8    182.2   -364.4     3467
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -3.5847 -0.6068 -0.0579  0.5103  4.3756
##
## Random effects:
##  Groups     Name               Variance Std.Dev. Corr
##  UserId     (Intercept)        0.002158 0.04645
##             agreementincorrect 0.001731 0.04160  0.10
##  set_number (Intercept)        0.002860 0.05348
##             agreementincorrect 0.002155 0.04642  -0.33
##  Residual                      0.049515 0.22252
## Number of obs: 3477, groups:  UserId, 88; set_number, 40
##
## Fixed effects:
##                      Estimate Std. Error         df t value Pr(>|t|)
## (Intercept)        -2.111e-02  1.666e-02  2.488e+02  -1.267 0.206287
## surprisal_value     1.472e-03  9.556e-04  3.143e+03   1.540 0.123604
## agreementincorrect  4.114e-02  1.160e-02  4.563e+01   3.546 0.000918 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr) srprs_
## surprisl_vl -0.743
## agrmntncrrc -0.113 -0.173
```

```
saveRDS(best_model8, file='PATH/3 - SPR main/analysis/best_model8.Rdata')
```

Permutation test

```
best_model8 <- readRDS('PATH/3 - SPR main/analysis/best_model8.Rdata')
perm8 <- permute_lmer(best_model8,
                      subset(data, word_index==8),
                      n_permutations,
                      verbose=TRUE)
saveRDS(perm8, file='PATH/3 - SPR main/analysis/perm8.Rdata')
```

```
plots8 <- plot_permutation_densities(perm8)
print(plots8)
```




```
print(perm8$p_values)
```

```
##        (Intercept)    surprisal_value agreementincorrect
##              0.447              0.098              0.000
```